

Assignment : Normalisation

Course Code: UQ25CA653A
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Course Title: Databases and Applications
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Question 1

A table **Student** is given:

StudentID	Name	Phone Numbers
S01	Megha R	98765..., 98450...
S02	Pranav S	90352..., 94488...

- Convert the table into **1NF**, **2NF**, and **3NF**.
- Clearly identify the **functional dependencies** and **primary keys** used at each step.

Question 2

Consider the relation:

$$R(A, B, C, D, E)$$

with the following functional dependencies:

$$\{ A \rightarrow BC, C \rightarrow D, B \rightarrow E \}$$

- Find all **candidate keys**.
- Determine the highest normal form satisfied by R.
- If it is not in 3NF, decompose until it becomes **3NF**.

Question 3

The following relation **Course** is given:

CourseID	CourseName	Faculty	Department
CS101	DSA	Anita K	CS
CS101	DSA	Anita K	Computing
CS102	DBMS	Ravi M	CA

- Identify any **redundancies** or **anomalies**.
- Perform **1NF**, **2NF**, and **3NF** conversions.
- Provide the **final decomposed schema**.

Question 4

Given the relation:

$$R(Student, Course, Instructor, Room)$$

with functional dependencies:

$$\{ Student, Course \rightarrow Instructor; Instructor \rightarrow Room \}$$

- Identify **partial** and **transitive** dependencies.
- Determine the highest normal form.
- Decompose into **BCNF** if required.

Question 5

Consider:

$$R(EmpID, EmpName, ProjID, ProjName, Manager)$$

with functional dependencies:

$$\{ EmpID \rightarrow EmpName, ProjID \rightarrow ProjName, ProjName \rightarrow Manager \}$$

- Identify all candidate keys.
- Check if the relation is in **1NF**, **2NF**, **3NF**, and **BCNF**.
- Decompose into the highest possible normal form.

Question 6

Given:

$$R(A, B, C, D, E)$$

with:

$$A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow E$$

- Identify the dependency chain.
- State whether the relation violates 3NF.
- Decompose into **3NF**.

Question 7

Hospital database table:

PatientID	PatientName	Doctor	DoctorPhone
P01	Arjun K	Dr. Mehta	99887...
P02	Riya S	Dr. Mehta	99887...
P03	Kiran V	Dr. Stella	88776...

- Find functional dependencies.
- Discuss insertion, update, deletion anomalies.
- Normalize the table up to **BCNF**.

Question 8

A table Student is given:

StudentID	Name	Phone Numbers
S01	Megha R	98765..., 98450...
S02	Pranav S	90352..., 94488...

Task: Convert this table into 1NF.

Question 9

Orders table:

OrderID	Customer	Items Ordered
O101	Ravi	Pen, Pencil, Eraser
O102	Roshni	Notebook, Marker

Task: Rewrite the table in 1NF ensuring atomic values.

Question 10

Course_Enrollment table:

StudentID	CourseID	StudentName	CourseName
S1	C1	Asha	DBMS
S1	C2	Asha	OS
S2	C1	Rohan	DBMS

Functional Dependencies:

- StudentID $\rightarrow$ StudentName
- CourseID $\rightarrow$ CourseName, Instructor
- (StudentID, CourseID) $\rightarrow$ all

Task: Convert to 2NF by removing partial dependencies.

Question 10

Employee_Project table:

EmpID	EmpName	Dept	Project	ProjectLocation
E1	Riya	IT	P1	Mumbai
E2	Karan	HR	P2	Delhi
E3	Riya	IT	P3	Mumbai

FDs:

- $\text{EmpID} \rightarrow \text{EmpName}, \text{Dept}$
- $\text{Project} \rightarrow \text{ProjectLocation}$

Task: Convert the relation to 3NF by removing transitive dependencies.

Question 11

Supplier_Parts table:

Supplier	City	Part	Cost
S1	Mumbai	P1	500
S1	Mumbai	P2	300
S2	Delhi	P1	500

FDs:

- $\text{Supplier} \rightarrow \text{City}$
- $(\text{Supplier}, \text{Part}) \rightarrow \text{Cost}$

Task: Is this table in BCNF? If not, decompose it.

Question 12

Orders_Detail table:

OrderID	CustID	CustomerName	ProductID	ProductName	Qty
O1	C1	Rahul	P1	Phone	1
O1	C1	Rahul	P2	Charger	2
O2	C2	Anitha	P1	Phone	1

FDs:

- $\text{CustID} \rightarrow \text{CustomerName}$
- $\text{ProductID} \rightarrow \text{ProductName}$
- $(\text{OrderID}, \text{ProductID}) \rightarrow \text{Qty}$

Task: Perform all steps of normalisation:

1. Convert to 1NF
2. Remove partial dependencies (2NF)
3. Remove transitive dependencies (3NF)
4. Check and convert to BCNF